

```
# =====
# SPEECH RECOGNITION SYSTEM
# Google Colab Project
# =====

# Install required libraries
!pip install SpeechRecognition pydub -q

import speech_recognition as sr
from google.colab import files
from pydub import AudioSegment
import os

# =====
# UPLOAD AUDIO FILE
# =====

print("Upload an audio file (WAV or MP3)")

uploaded = files.upload()

audio_file = list(uploaded.keys())[0]

print(f"\nUploaded File: {audio_file}")

# =====
# CONVERT MP3 TO WAV IF NEEDED
# =====

if audio_file.endswith('.mp3'):

    print("\nConverting MP3 to WAV...")

    sound = AudioSegment.from_mp3(audio_file)

    wav_file = "converted_audio.wav"

    sound.export(wav_file, format="wav")

else:
    wav_file = audio_file

# =====
```

```
# INITIALIZE RECOGNIZER
# =====

recognizer = sr.Recognizer()

# =====
# LOAD AUDIO FILE
# =====

with sr.AudioFile(wav_file) as source:

    print("\nReading audio file...")

    audio_data = recognizer.record(source)

# =====
# SPEECH TO TEXT
# =====

try:

    print("\nConverting Speech to Text...\n")

    text = recognizer.recognize_google(audio_data)

    print("=====")
    print("RECOGNIZED TEXT")
    print("=====\n")

    print(text)

    print("\n=====")
    print("CONVERSION SUCCESSFUL")
    print("=====")

except sr.UnknownValueError:

    print("Could not understand the audio.")

except sr.RequestError as e:

    print(f"Error connecting to Google Speech API: {e}")

# =====
# SAVE OUTPUT TO TXT FILE
```

```
# =====

save_option = input("\nDo you want to save text to file? (y/n): ")

if save_option.lower() == 'y':

    with open("recognized_text.txt", "w") as f:

        f.write(text)

    files.download("recognized_text.txt")

    print("\nrecognized_text.txt downloaded successfully!")
```

32.9/32.9 MB 34.5 MB/s eta 0:00:00

Upload an audio file (WAV or MP3)

```
/usr/local/lib/python3.12/dist-packages/pydub/utils.py:300: SyntaxWarning: inv
m = re.match('([su]([0-9]{1,2})p?) \(([0-9]{1,2}) bit\)$', token)
/usr/local/lib/python3.12/dist-packages/pydub/utils.py:301: SyntaxWarning: inv
m2 = re.match('([su]([0-9]{1,2})p?) ( \(default\) )?$', token)
/usr/local/lib/python3.12/dist-packages/pydub/utils.py:310: SyntaxWarning: inv
elif re.match('(flt)p? ( \(default\) )?$', token):
/usr/local/lib/python3.12/dist-packages/pydub/utils.py:314: SyntaxWarning: inv
elif re.match('(dbl)p? ( \(default\) )?$', token):
```

freesound_...h-94649.mp3

freesound_community-speech-94649.mp3(audio/mpeg) - 48588 bytes, last modified: 12/05/2026 - 100% done

Saving freesound_community-speech-94649.mp3 to freesound_community-speech-9464

Uploaded File: freesound_community-speech-94649.mp3

Converting MP3 to WAV...

Reading audio file...

Converting Speech to Text...

```
=====
RECOGNIZED TEXT
=====
```

you are just a line of code

```
=====
CONVERSION SUCCESSFUL
=====
```

Do you want to save text to file? (y/n): y

Start coding or [generate](#) with AI.

